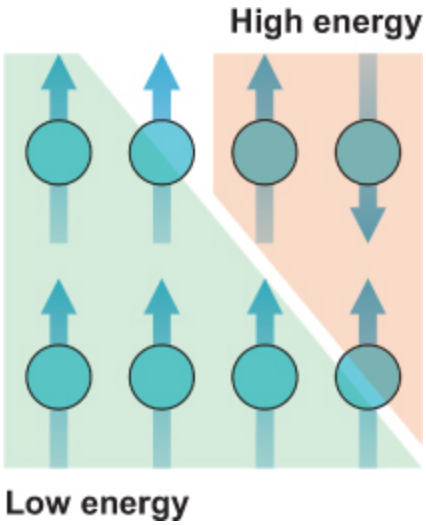
iron can become a big magnet if all of its electrons are aligned in the same direction (i.e., all spin-up or all spin-down).

Physicists were trying to study how the electrons “decide” to align themselves, and how the temperature of the iron affects this process. If you heat up a magnet, at some point the aligned electrons will start randomly alternating their spins so that the material loses its net magnetic field. Physicists knew that an individual electron creates a magnetic field, and that a tiny magnetic field will affect a nearby electron. If you’ve ever played with two bar magnets, you’ve noticed that they will naturally line up in one direction or repel in the opposite direction. The electrons do the same thing. It makes sense that electrons would also try to align themselves to be the same spin (figure 9.9).

Figure 9.9. When electrons are packed together, they prefer to have their spins aligned in the same direction because it is a lower energy configuration than when their spins are anti-aligned, and all physical systems tend toward lower energy (all else being equal).



There’s one added complexity though. Although individual electrons have a tendency to align themselves, a sufficiently large group of aligned electrons actually becomes unstable. This is because as the number of aligned electrons grows larger, the magnetic field grows and creates some internal strain on the material. So what really happens is that electrons will form clusters, called *domains*, in which all the electrons are aligned (either spin up or down), but other domains also form. For example, there might be a